Project: Library Management System

Objective:-

To build a complete book and library management system with book and member classes implementing proper OOP principles,encapsulation and basic library operations.

Project Structure:-

Books.java:- *Book class with attributes: ISBN, title, author, genre, availability .*

Member.java:- *Member class with attributes: memberId, name, contact, borrowedBooks .*

LibraryManagementSystem:- *Class to implement the complete functionality*

addBook()- *function to add a book to the list of books.*
addmember()- *function to add a member to the list of members.*
displayAllBooks()- *function to display all books available in the library.*
displayAvailableBooks()- *function to display the books only which are available for borrowing.*
searchBooks() - *search a book based on a keyword.*
borrowbook()- *enables a member to borrow a book.*
returnbook()- *enables a member to return a book.*

Code:-

Books Class:

```
public class Books {
	private String isbn;
	private String title;
	private String author;
	private String genre;
	private boolean availablilty;

	//Parameterized constructor
			public Books(String isbn, String title, String author, String genre) {
				super();
				this.isbn = isbn;
				this.title = title;
```

```
                this.author = author;
                this.genre = genre;
                this.availablilty = true;
        }

        //Getters and Setters
        public String getIsbn() {
                return isbn;
        }
        public void setIsbn(String isbn) {
                this.isbn = isbn;
        }
        public String getTitle() {
                return title;
        }
        public void setTitle(String title) {
                this.title = title;
        }
        public String getAuthor() {
                return author;
        }
        public void setAuthor(String author) {
                this.author = author;
        }
        public String getGenre() {
                return genre;
        }
        public void setGenre(String genre) {
                this.genre = genre;
        }
        public boolean isAvailable() {
                return availablilty;
        }
        public void setAvailablilty(boolean availablilty) {
                this.availablilty = availablilty;
        }

        //Display details
        public void displaydetails() {
                System.out.println("ISBN :"+isbn);
                System.out.println("Title :"+title);
                System.out.println("Author :"+author);
                System.out.println("Genre :"+ genre);
                System.out.println("Availability :"+(availablilty ? "Yes":"No"));
        }

}
```

Member Class:

```
import java.util.*;
//Member class- to contain member detail and borrowing and returning functionality
public class Member {
 private String memberId;
 private String name;
 private String contact;
 private List<Books> Borrowed;

 //Parameterized constructor
 public Member(String memberId,String name,String contact) {
        super();
        this.memberId = memberId;
        this.name = name;
        this.contact = contact;
        Borrowed = new ArrayList<>();
 }

 //Getters and Setters
 public String getMemberId() {
        return memberId;
 }
 public void setMemberId(String memberId) {
        this.memberId = memberId;
 }
 public String getName() {
        return name;
 }
 public void setName(String name) {
        this.name = name;
 }
 public String getContact() {
        return contact;
 }
 public void setContact(String contact) {
        this.contact = contact;
 }
 public List<Books> getBorrowed() {
        return Borrowed;
 }

 //Book borrowing functionality
 public void borrowBook(Books book) {
```

```
            if(book.isAvailable()) {
                    Borrowed.add(book);
                    book.setAvailablilty(false);
                    System.out.println("Book has been borrowed succesfully");
            }
            else {
                    System.out.println("We are sorry! Book is currently unavailable");
            }
 }

 //Book returning functionality
 public void returnBook(Books book) {
            Borrowed.remove(book);
            book.setAvailablilty(true);
            System.out.println("Book returned succesfully");
}
]
```

Library management System Class:

```
import java.util.*;
//Main class
public class LibraryManagementSystem {
  static Scanner sc=new Scanner(System.in);

  //List to store the books
  List<Books> books=new ArrayList<>();

  //List to store the members
  List<Member> member=new ArrayList<>();

  //Add member functionality
  void addmember() {
          System.out.println("\n===ADD MEMBER===\n");
          System.out.println("-".repeat(100));
          System.out.println("Enter member Id");
          sc.nextLine();
          String id=sc.nextLine();
          System.out.println("Enter member name");
          String name=sc.nextLine();
          System.out.println("Enter member contact");
          String contact=sc.nextLine();
          member.add(new Member(id,name,contact));
          System.out.println("-".repeat(100));
  }
```

```
//Add book functionality
 void addBook() {
        System.out.println("\n===ADD BOOK===\n");
        System.out.println("-".repeat(100));
        System.out.println("Enter ISBN no.");
        sc.nextLine();
        String isbn=sc.nextLine();
        System.out.println("Enter Book title");
        String title=sc.nextLine();
        System.out.println("Enter Book author");
        String author=sc.nextLine();
        System.out.println("Enter Book genre");
        String genre=sc.nextLine();
        books.add(new Books(isbn,title,author,genre));
        System.out.println("-".repeat(100));
 }

 //Display books
 void displayAllBooks() {
        System.out.println("\n===DISPLAY BOOKS===\n");
        System.out.println("-".repeat(100));
        for(Books book:books) {
                book.displaydetails();
        }
        System.out.println("-".repeat(100));
 }

 //Display available books
 void displayAvailableBooks() {
        System.out.println("\n===DISPLAY AVAILABLE BOOKS===\n");
        System.out.println("-".repeat(100));
        for(Books book:books) {
                if(book.isAvailable()) {
                        book.displaydetails();
                }
        }
        System.out.println("-".repeat(100));
 }

 //Search books by a keyword
 void searchBooks() {
        System.out.println("\n===SEARCH BOOKS===\n");
        System.out.println("-".repeat(100));
       ArrayList<Books> results = new ArrayList<>();
       System.out.println("Enter keyword");
       sc.nextLine();
```

```
        String keyword=sc.nextLine();
        keyword = keyword.toLowerCase();
        for(Books book : books) {
           if(book.getTitle().toLowerCase().contains(keyword) ||
              book.getAuthor().toLowerCase().contains(keyword) ||
              book.getGenre().toLowerCase().contains(keyword)) {
               results.add(book);
           }
        }
        System.out.println("List of Books that matches your search are:");
        for(Books book: results) {
           book.displaydetails();
     }
        System.out.println("-".repeat(100));
}

//Borrow books
void borrowbook() {
          System.out.println("\n===BORROW BOOK===\n");
          System.out.println("-".repeat(100));
          System.out.println("Enter your memberId");
          sc.nextLine();
          String memberId=sc.nextLine();
          System.out.println("Enter isbn number of book");
          String isbn=sc.nextLine();
          Member borrower=null;
          Books toBorrow=null;
          for(Member m:member) {
                    if(m.getMemberId().equalsIgnoreCase(memberId)) {
                              borrower=m;
                    }
          }
          for(Books b:books) {
                    if(b.getIsbn().equalsIgnoreCase(isbn)) {
                              toBorrow=b;
                    }
          }
          if((borrower!=null)&&(toBorrow!=null)) {
          borrower.borrowBook(toBorrow);
          }
          else {
                    System.out.println("Invalid input!!!");
          }
          System.out.println("-".repeat(100));
}

//Return books
```

```
void returnbook() {
	System.out.println("\n===RETURN BOOK===\n");
	System.out.println("-".repeat(100));
	System.out.println("Enter your memberId");
	sc.nextLine();
	String memberId=sc.nextLine();
	System.out.println("Enter isbn number of book");
	String isbn=sc.nextLine();
	Member returner=null;
	Books toReturn=null;
	for(Member m:member) {
		if(m.getMemberId().equalsIgnoreCase(memberId)) {
			returner=m;
		}
	}
	for(Books b:books) {
		if(b.getIsbn().equalsIgnoreCase(isbn)) {
			toReturn=b;
		}
	}
	List<Books> borrowed=returner.getBorrowed();
	if((returner!=null)&&(toReturn!=null)&&(borrowed.contains(toReturn))) {
	returner.returnBook(toReturn);
	}
	else {
	System.out.println("Inavlid entry!!!");
	}
	System.out.println("-".repeat(100));
}

//Main function
public static void main(String args[]) {
	int choice=0;
	System.out.println("\n===Library Management System===\n");
	LibraryManagementSystem lms= new LibraryManagementSystem();
	while(choice!=8) {
		System.out.println("1. Add New Book\r\n"
				+ "2. Register New Member\r\n"
				+ "3. Display All Books\r\n"
				+ "4. Display Available Books\r\n"
				+ "5. Search Books\r\n"
				+ "6. Borrow Book\r\n"
				+ "7. Return Book\r\n"
				+ "8. Exit\r\n"
				+ "Enter your choice:");
		choice=sc.nextInt();
		if(choice==1) {
```

```
                        lms.addBook();
                }
                else if(choice==2) {
                        lms.addmember();
                }
                else if(choice==3) {
                        lms.displayAllBooks();
                }
                else if(choice==4) {
                        lms.displayAvailableBooks();
                }
                else if(choice==5) {
                        lms.searchBooks();
                }
                else if(choice==6) {
                        lms.borrowbook();
                }
                else if(choice==7) {
                        lms.returnbook();
                }
        }
  }

}


 }


}
```

OUTPUT:-

===Library Management System===

1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit

Enter your choice:
1

===ADD BOOK===

Enter ISBN no.
11
Enter Book title
Harry Potter
Enter Book author
JK Rowling
Enter Book genre
Sci-fi
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
1

===ADD BOOK===

Enter ISBN no.
12
Enter Book title
Frankestein
Enter Book author
Marry Shelly
Enter Book genre
Horror
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
1

===ADD BOOK===

Enter ISBN no.
13
Enter Book title

The Merchant Of Venice
Enter Book author
William Shakespeare
Enter Book genre
Drama
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
2

===ADD MEMBER===

Enter member Id
101
Enter member name
Bob
Enter member contact
779773
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
2

===ADD MEMBER===

Enter member Id
102
Enter member name
Alice
Enter member contact
81673
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books

5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
2

===ADD MEMBER===

Enter member Id
103
Enter member name
Shilpa
Enter member contact
81673
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
3

===DISPLAY BOOKS===

ISBN :11
Title :Harry Potter
Author :JK Rowling
Genre :Sci-fi
Availability :Yes
ISBN :12
Title :Frankestein
Author :Marry Shelly
Genre :Horror
Availability :Yes
ISBN :13
Title :The Merchant Of Venice
Author :William Shakespeare
Genre :Drama
Availability :Yes
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books

6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
5

===SEARCH BOOKS===

Enter keyword
Frank
List of Books that matches your search are:
ISBN :12
Title :Frankestein
Author :Marry Shelly
Genre :Horror
Availability :Yes
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
6

===BORROW BOOK===

Enter your memberId
101
Enter isbn number of book
12
Book has been borrowed succesfully
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
4

===DISPLAY AVAILABLE BOOKS===

ISBN :11
Title :Harry Potter
Author :JK Rowling
Genre :Sci-fi
Availability :Yes
ISBN :13
Title :The Merchant Of Venice
Author :William Shakespeare
Genre :Drama
Availability :Yes
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
7

===RETURN BOOK===

Enter your memberId
101
Enter isbn number of book
12
Book returned succesfully
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice:
4

===DISPLAY AVAILABLE BOOKS===

ISBN :11
Title :Harry Potter
Author :JK Rowling
Genre :Sci-fi
Availability :Yes
ISBN :12
Title :Frankestein

Author :Marry Shelly
Genre :Horror
Availability :Yes
ISBN :13
Title :The Merchant Of Venice
Author :William Shakespeare
Genre :Drama
Availability :Yes
---
1. Add New Book
2. Register New Member
3. Display All Books
4. Display Available Books
5. Search Books
6. Borrow Book
7. Return Book
8. Exit
Enter your choice: